Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 07 · Part B: Mathematics of Signals and Systems Level 2

Track: Random Processes, Autocorrelation, Power Spectral Density, Colored Noise, Photon Counting Statistics

Prerequisites: Problems 01, 04 · **Estimated effort:** 12–16 hours

Random Signals, Noise and Power Spectral Density: The Mathematics of Not Knowing

Noise is not "junk to be removed"; it is a random process with a measurable correlation and spectrum, and every detection threshold is a statement about its tail. Estimate autocorrelations and spectra, shape noise with filters, generate and measure $1/f$ noise, and discover why a 3σ threshold on photon counts is not a 3σ threshold at all.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/07_random_signals_and_noise_spectra/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random

seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis sections, and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

Every instrument record is a single realisation of a random process. To say that a feature is "significant" you must know the statistics of the background: its distribution, how correlated it is from one sample to the next, and how its power is spread across frequency. White Gaussian noise averages away like $1/\sqrt{N}$; correlated noise does not. A gamma-ray detector counts photons whose arrival times are Poisson, so its noise depends on the signal level itself, and at low count rates the Gaussian approximation underestimates the false-alarm rate by large factors.

This problem builds the statistical vocabulary used by every later problem on filtering, detection and machine learning. The central objects are the autocorrelation, its Fourier transform (the power spectral density), and the Poisson distribution.

Theoretical Framework

1. Random Variables and Processes

A random process $x[n]$ is *wide-sense stationary* if its mean and its autocorrelation $R_x[m] = \mathbb{E}\{x[n]x[n+m]\}$ do not depend on n; it is *ergodic* if time averages over one long record converge to ensemble averages. For white noise of variance σ^2, $R_x[m] = \sigma^2\delta[m]$. By the central limit theorem sums of many independent contributions tend to a Gaussian.

2. Power Spectral Density

The Wiener–Khinchin theorem states that the PSD is the Fourier transform of the autocorrelation,

$$S_x(f) = \sum_{m=-\infty}^{\infty} R_x[m]\, \mathrm{e}^{-\mathrm{j}2\pi f m/f_s}\, \frac{1}{f_s}, \qquad \sigma_x^2 = R_x[0] = \int_{-f_s/2}^{f_s/2} S_x(f)\, \mathrm{d}f, \tag{1}$$

(two-sided, in units^2/Hz). White noise has the flat two-sided density σ^2/f_s, i.e. the one-sided density $2\sigma^2/f_s$. Passing a random signal through an LTI system H gives $S_y(f) = |H(f)|^2 S_x(f)$; for white input the output variance is $\sigma^2 \sum_n h[n]^2$ and the equivalent noise bandwidth is $\mathrm{ENBW} = \int |H|^2 \mathrm{d}f / |H_{\max}|^2$.

3. AR(1) Process and Effective Sample Size

$x[n] = a\,x[n-1] + w[n]$ with white w of variance σ_w^2 has

$$R_x[m] = \frac{\sigma_w^2}{1-a^2}a^{|m|}, \qquad S_x(f) = \frac{\sigma_w^2/f_s}{|1 - a\mathrm{e}^{-\mathrm{j}2\pi f/f_s}|^2}. \tag{2}$$

The mean of N samples of such a correlated process has variance of an effective number of independent samples $N_{\text{eff}} \approx N(1-a)/(1+a)$.

4. $1/f$ Noise

Flicker noise has $S(f) \propto 1/f^{\alpha}$ with $\alpha \approx 1$; it can be synthesised by multiplying the spectrum of white noise by $f^{-\alpha/2}$. It does not average down (problems 01 and 09) and appears as a flat Allan deviation.

5. Poisson Counting

Photon (or event) counts in a bin of duration Δt at rate λ are Poisson distributed with mean and variance both equal to $\mu = \lambda\Delta t$. The variance-to-mean ratio (the Fano factor) is 1; a Fano factor above 1 indicates extra variability (drift, bursts, clustering). For large μ the Poisson tends to a Gaussian with $\sigma = \sqrt{\mu}$, but its upper tail is heavier. For a source of N_s counts over a background of N_b counts the signal-to-noise ratio is

$$\mathrm{S/N} = \frac{N_s}{\sqrt{N_s + N_b}}. \tag{3}$$

Data and Simulation Specification

Distributions	Gaussian, uniform, Laplace and Poisson ($\mu = 2, 10, 100$), 10^5 samples; CLT test: sums of $1, 2, 4, 8, 16$ uniforms.
AR(1)	$a = 0.0$, 0.5, 0.9, 0.99; $\sigma_w = 1$; $N = 2^{16}$; 500 realisations for ensemble averages.
$1/f$ **noise**	Spectral exponent $\alpha = 1$, $f_s = 100\,\mathrm{Hz}$, $N = 2^{18}$.
Counts (synthetic)	`sp_common.synth.grb_event_list(seed=7)`: event times of a burst on a drifting background (250 Hz nominal); bin at 0.1 s, 1 s, 10 s; use the interval $t < 100\,\mathrm{s}$ as "background only".
Counts (real, choose one)	Fermi GBM time-tagged event data for a quiet interval or a burst (HEASARC), or an AstroSat CZTI level-2 event file; a TESS or Kepler light curve for photon-noise-limited flux.
Tail probabilities	Poisson with mean 5 versus Gaussian $N(5,5)$ at the $+3\sigma$ threshold; Gaussian one-sided 3σ and 5σ for 1 Hz bins over one day.

Your Task

1. **Distributions and the CLT (Stage 2).** Sample the four distributions, verify mean and variance and plot histograms against the analytic densities. Show the sum of k uniforms approaching a Gaussian via skewness and excess kurtosis and a Q–Q plot.
2. **Autocorrelation estimators.** Implement the biased $(1/N)$ and unbiased $(1/(N-m))$ autocorrelation estimators. Compare white noise and AR(1) with the theory of Eq. (2). Demonstrate ergodicity by comparing the time-average estimate of $R_x[1]$ from one long record with the ensemble average over 500 short records, and show a non-ergodic counterexample (random constant offset per realisation).

3. **Wiener–Khinchin.** Compute the PSD as the Fourier transform of the estimated autocorrelation and as the periodogram $|X[k]|^2/(Nf_s)$. Verify $\int S\,\mathrm{d}f = \sigma^2$ and the one-sided scaling factor 2. Show that the periodogram's scatter does *not* shrink as N grows and state its distribution (preview of problem 13).
4. **Filtering random signals.** Pass white noise through a moving average and through a first-order IIR. Verify $S_y = |H|^2 S_x$, compute the output variance and the ENBW, and check the $1/M$ variance reduction for a moving average of white noise and the N_{eff} rule for AR(1) data.
5. **Colored noise.** Synthesise $1/f$ noise by spectral shaping and Brownian ($1/f^2$) noise by integration. Estimate the spectral slope on a log–log plot with a fitted line and confidence interval; compute the Allan deviation of each and relate its slope to the spectrum.
6. **Poisson statistics.** Simulate counts for $\mu = 2, 10, 100$, compare histograms with the Poisson mass function and the Gaussian approximation, and compute the probability of exceeding $\mu + 3\sqrt{\mu}$ in both. Quantify for $\mu = 5$ the ratio of the true false-alarm probability to the Gaussian prediction.
7. **Real or synthetic counts.** Bin the event list at three widths. In the burst-free interval compute the Fano factor at each width; identify over-dispersion caused by the slow background drift and explain why it motivates detrending (problem 18). Compute S/N of the burst with Eq. (3).

Self-check targets

For AR(1) with $a = 0.9$: $R_x[1]/R_x[0] = 0.9$, $N_{\text{eff}} = N/19$, and $S_x(0) = \sigma_w^2/(f_s(1-a)^2)$. For a moving average of M white samples the variance drops by exactly $1/M$. For $\mu = 5$ the Poisson probability of $N \geq 12$ (the $+3\sigma$ threshold is 11.7) is 0.0055, about 4 times the Gaussian 0.00135. A one-sided 3σ threshold on 1 Hz bins gives about 117 false alarms per day, 5σ about 0.025 per day.

Aerospace Application Link

Gamma-ray burst detectors trigger on count-rate excesses over a background of a few hundred counts per second. For a background of 266 counts per 1 s bin, the Gaussian 3σ threshold is ≈ 314 counts but the true Poisson false-alarm probability at that threshold differs from 0.00135. Using your Task 6–7 results, find the threshold (in counts per bin) that gives at most one false trigger per day for a search over 1-second bins, state the minimum burst fluence (extra counts) that is detectable at that threshold, and explain why a multi-timescale search (several bin widths) raises the number of independent trials.

Deliverables

D1 **Distribution study:** histograms, moments and the CLT demonstration with Q–Q plots.

D2 **Correlation and PSD study:** estimators, ergodicity test, Wiener–Khinchin verification with scaling checks.

D3 **Filtered noise:** verification of $S_y = |H|^2 S_x$, ENBW, variance reduction and N_{eff} for AR(1).

D4 **Colored noise:** $1/f$ and Brownian synthesis, fitted slopes with confidence intervals, Allan deviations.

D5 **Poisson study:** histograms, tail probabilities, the Fano-factor table and the count significance calculation.

D6 **Memo (250 words):** "Why is a 3σ threshold on photon counts not a 3σ threshold?" Cite at least two external sources.

Bonus (a) Compute the exact distribution of the maximum of N Gaussian samples and compare with Monte Carlo. (b) Derive the periodogram's χ^2_2 distribution. (c) Implement the Li–Ma significance for on/off counting and compare with Eq. (3).

Learning Resources

Random Signals and Spectra

- Papoulis, A., Pillai, S.U. *Probability, Random Variables and Stochastic Processes*, chapters on stochastic processes and spectral representation.
- Oppenheim, A.V., Schafer, R.W. *Discrete-Time Signal Processing*, Ch. 2.10 and Ch. 10 (random signals and power spectrum estimation).
- Kay, S.M. *Modern Spectral Estimation: Theory and Application* (AR models and spectral density).

Counting Statistics and Astronomy Data

- Li, T.-P., Ma, Y.-Q. (1983). "Analysis methods for results in gamma-ray astronomy." Astrophys. J. 272, 317.
- Fermi GBM data via HEASARC: `https://heasarc.gsfc.nasa.gov`; AstroSat data via the ISSDC archive: `https://astrobrowse.issdc.gov.in/astro_archive/archive/Home.jsp`

Tools

- NumPy random generators and SciPy stats: `https://numpy.org/doc/stable/reference/random/`, `https://docs.scipy.org/doc/scipy/reference/stats.html`

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.